Strategy: Causal loop diagram

Fundamentals of the strategy

What is the goal of this strategy ?

Use **causal loop diagrams** to **visualize** and understand **complex interactions**, **impacts** and **feedback loops** in complex **systems**, in particular socio-technical systems.

What is system thinking ?

System thinking is an approach to understanding and managing complex systems by recognizing the **interconnections and interactions** between their various components, focusing on **the whole system** rather than individual parts to improve performance and solve issues effectively.

System

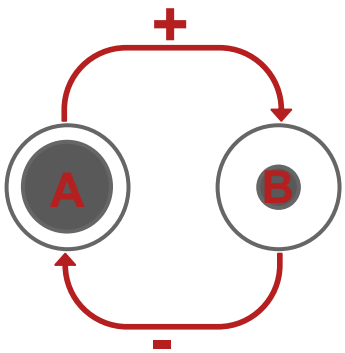
Collection of organized parts interacting with each other to function as a whole.

Loops

Balancing loops:

feedback loop that **maintains stability** within a system by counteracting changes. It brings the system back to a desired state or equilibrium whenever deviations occur.

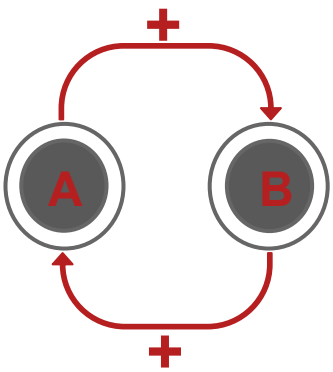
Example:
A ↗
B ↗
A ↘
"An increase leads to a decrease"



Reinforcing loops:

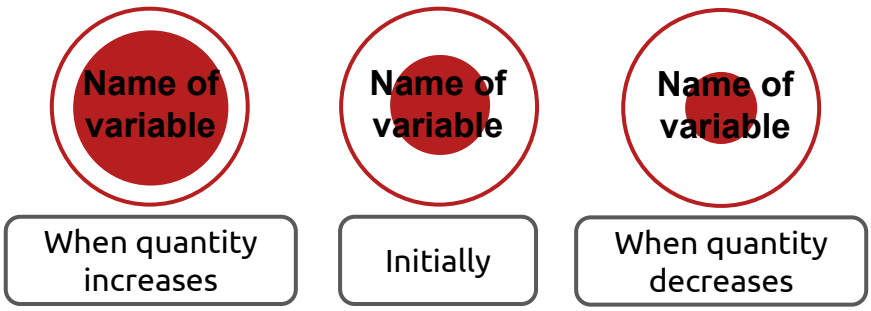
feedback loop that **amplifies changes** within a system, leading to **growth or decline** over time. It reinforces the direction of change, creating a cycle where an increase/decrease in one variable causes further increase/decrease.

Example:
A ↗
B ↗
A ↗
"An increase leads to a increase"

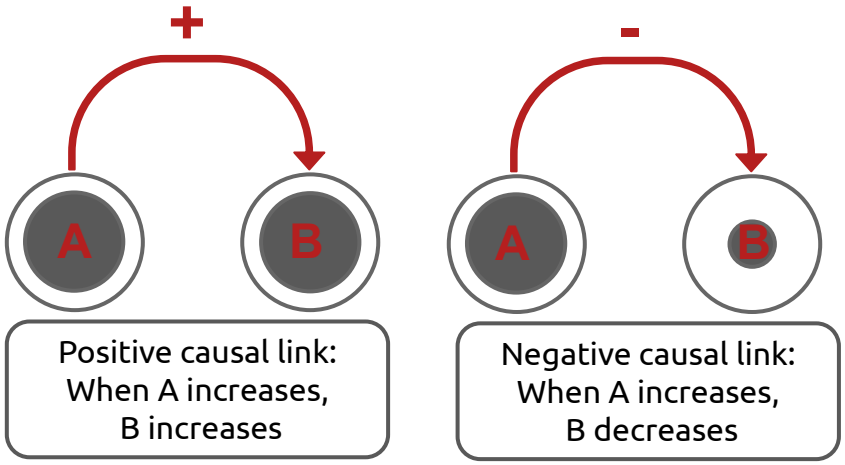


Causal loop diagrams elements

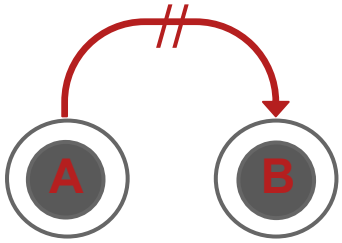
1. **Key variables: element** within a system that significantly influences the **behavior** and outcomes of the system. Each variable holds a **quantity** whose value **changes over time**.
It is represented this way:



2. **Causal link:** indicates how one variable influences another, showing the **direction** of the relationship: **positive = same direction, negative = opposite direction**. It is represented this way:



3. **Delayed effect:** the effect of one variable on another **does not occur immediately** but after some time (time lag in the system's response).



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